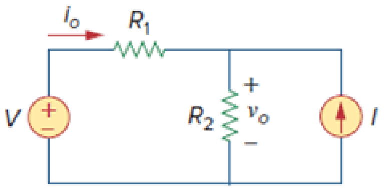
Chapter 4, Problem 21P

Problem

Using Fig. 4.89, design a problem to help other students better understand source transformation.

Figure. 4.89:



Step-by-step solution

Step 1 of 1

Using Fig. 4.89, design a problem to help other students better understand source transformation.

Although there are many ways to work this problem, this is an example based on the same kind of problem asked in the third edition.

Problem

Apply source transformation to determine ***vo*** and ***io*** in the circuit in Fig. 4.89.

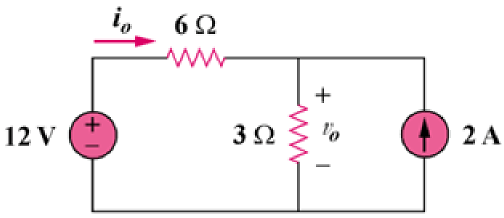
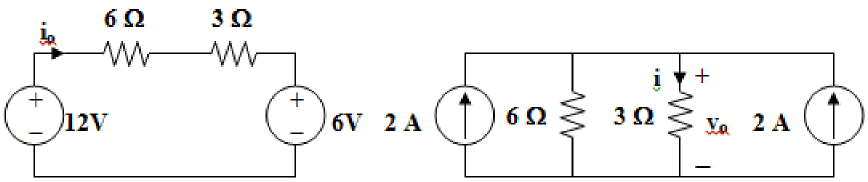


Figure 4.89

Solution

To get io, transform the current sources as shown in Fig. (a).



(a) (b)

From Fig. (a), $-12 + 9io + 6 = 0$, therefore $io = $ **666.7 mA**

To get vo, transform the voltage sources as shown in Fig. (b).

$i = [6/(3 + 6)](2 + 2) = 8/3$

$vo = 3i = $ **8 V**

Comments (2)

Anonymous

When it says -12 + 9io + 6 = 0, how do we know when the 12V is negative? The 9io is in the same direction (currentwise) so why wouldnt the 6V be negative?

Anonymous

Wait I got it, the 12V source is supplying power so it is negative